Self Average Approximation

Hypothesis:

In the model, an agent interacts with her neighbours and observe the clauses from the universe to add to her knowledge base. We also keep changing the universal clauses every few steps. Assuming, that there is a bigger pool of clauses ($\mathcal{C}$) from which we are bringing the clauses into the universe($\mathcal{U}$).
Since, the pool of clauses we are sampling from is a uniformly distributed pool of `XOR` and `AND` clauses, even though the clauses keep getting updated, the Knowledge base K_B for each agent will soon resemble the statistics of $\mathcal{C}$.

Consider the model in the long-time regime, after repeated private observation and communication have filled each agent's knowledge base with clauses sampled from the common clause pool.

Assumptions:

- clauses are sampled uniformly from the same global pool,
- the universe contains $M = \alpha K$ clauses,
- half the clauses are `AND` and half are `XOR`,
- Each clause is made up of two variables.

Theorem

Theorem. Under the mean-field approximation, the expected long-time violation count is minimized at $p^\star = 1$, and the corresponding attractor value for average violations is

$$V^* = \frac{\alpha K}{2}$$

This attractor depends on the clause density parameter α and the number of variables K, but not on the network topology, N, `obs_prob`, or `clause_interval`.

Proof sketch

Let p be the probability that a given variable equals 1 in the steady state. Each agent has K variables. So the state of variables that each agent owns is : $\vec{x} \in \{0,1\}^K$. Therefore, we can say p is the amount of 1 in the states for an agent.

A clause consists of two variables, so for an individual clause, we need to multiply the probability twice.

For an `AND` clause on two variables, satisfaction requires both variables to be 1, so

$P(\text{AND satisfied}) = p^2, \qquad P(\text{AND violated}) = 1 - p^2$

For an `XOR` clause, satisfaction requires exactly one variable to be 1, so

$P(\text{XOR satisfied}) = p(1-p) + (1-p)p = 2p(1-p)$ and
$P(\text{XOR violated}) = 1 - 2p(1-p) = p^2 + (1-p)^2$.

If half of the M clauses are `AND` and half are `XOR`, then the expected total violation count is

$V(p) = \frac{M}{2}(1-p^2) + \frac{M}{2}\left(p^2 + (1-p)^2\right)$

Simplifying,

$$V(p) = M\left(1 - p + \frac{p^2}{2}\right)$$

Differentiating gives

$$\frac{dV}{dp} = M(p-1)$$

so the only stationary point is $p^\star = 1$. The second derivative is

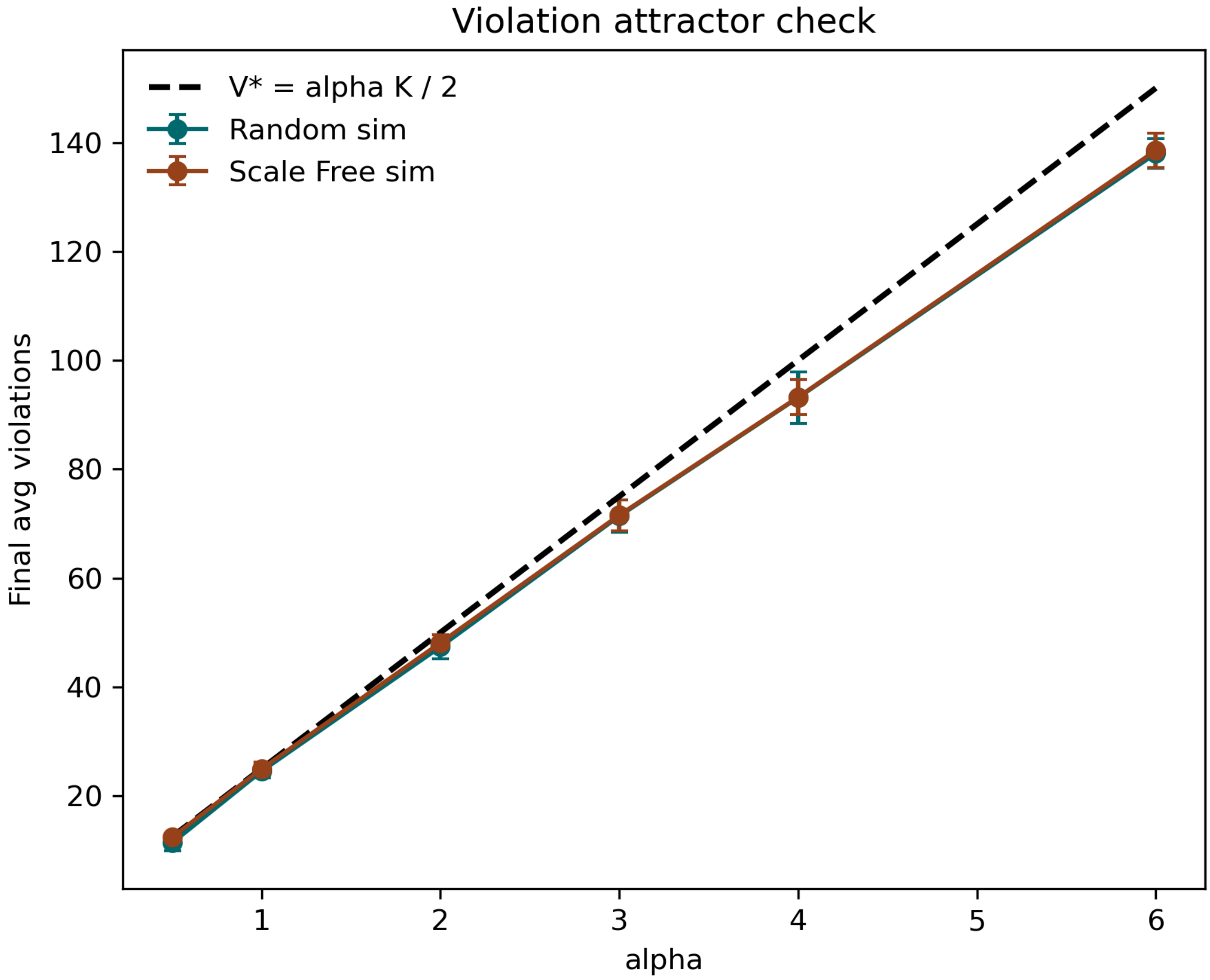
$\frac{d^2V}{dp^2} = M > 0$

so this point is a global minimum. Substituting $p^\star = 1$ yields

$$V^* = M/2 = \alpha K/2.$$

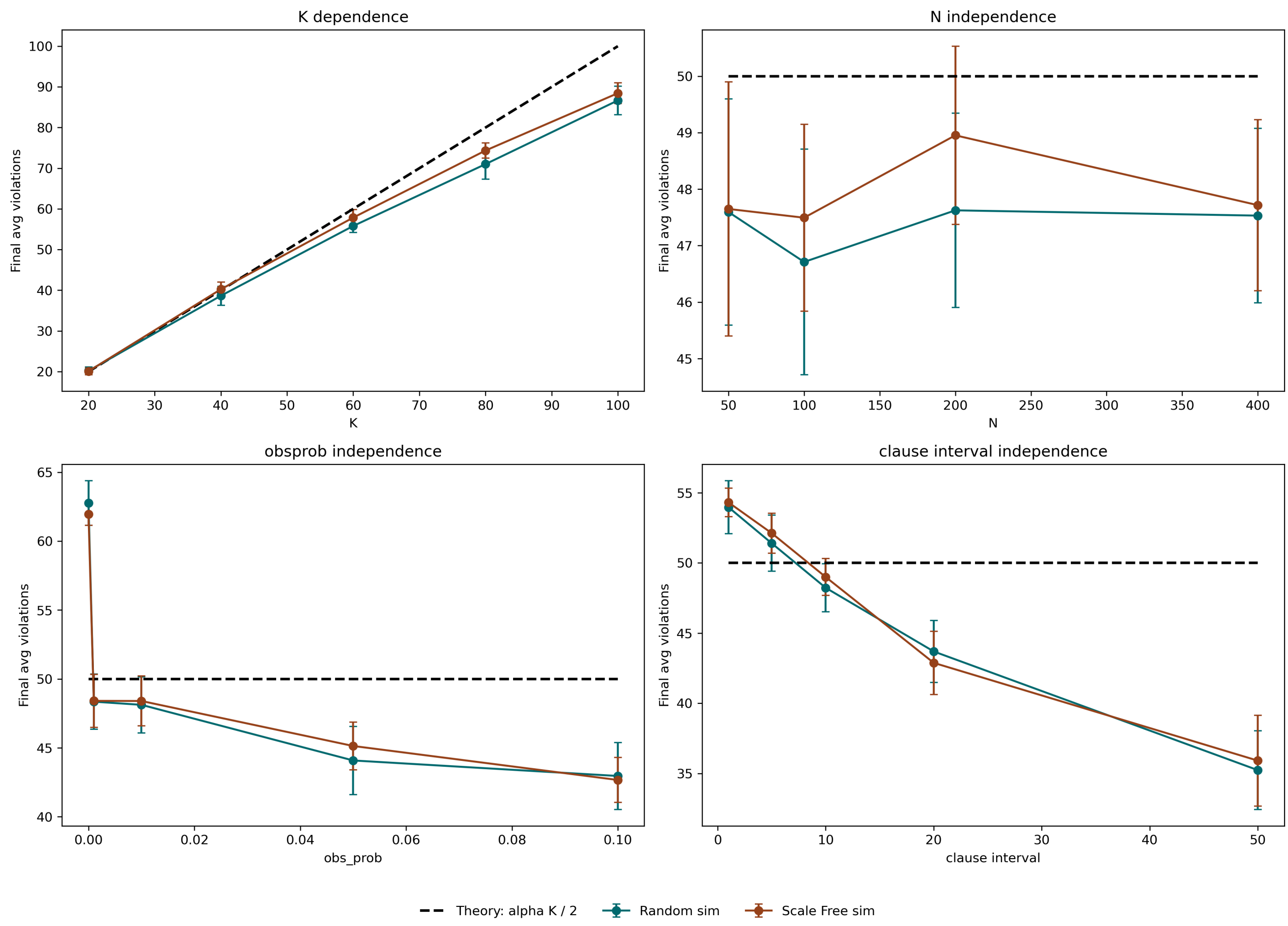
Verification

I ran the simulation for 10 erdos-renyi random graphs and 10 scale-free random graphs, I checked the average violations after 3000 graph updates varying α, keeping K constant. There is a clear correlation with the expected attractor value.



The following plot shows the correlation with similar experiment done for the attractor with K keeping α constant. Note also, that when we keep everything else constant and vary N, observation probability and Clause interval, the violations do not follow theoretical attractor.

There is a decline in violation ofcourse in increasing clause interval, as this is keeping the universe's clause distribution less dynamic and thus it is easier to reach violations.



Why network independence exists

The network does matter for transient mixing, but not for the final attractor, for what feels like two reasons.

First, `comm_scale` normalizes the expected incoming information flow by the average in-degree, so each agent receives roughly the same amount of communication on average regardless of whether the graph is Random, Scale Free, or another topology.

In other words, the topology changes *who* transmits information, but not the average *rate* at which information enters the local K_B.

Second, once communication and private observation have repeatedly sampled the common clause pool, every agent's K_B becomes a moving sample of the same global clause distribution.

At that point, the local greedy step sees essentially the same clause statistics everywhere, so the long-time objective is the same on all nodes. The network can speed up or slow down convergence, but it does not change the mean-field.